Rotary Wing Aerodynamics (AE 667)



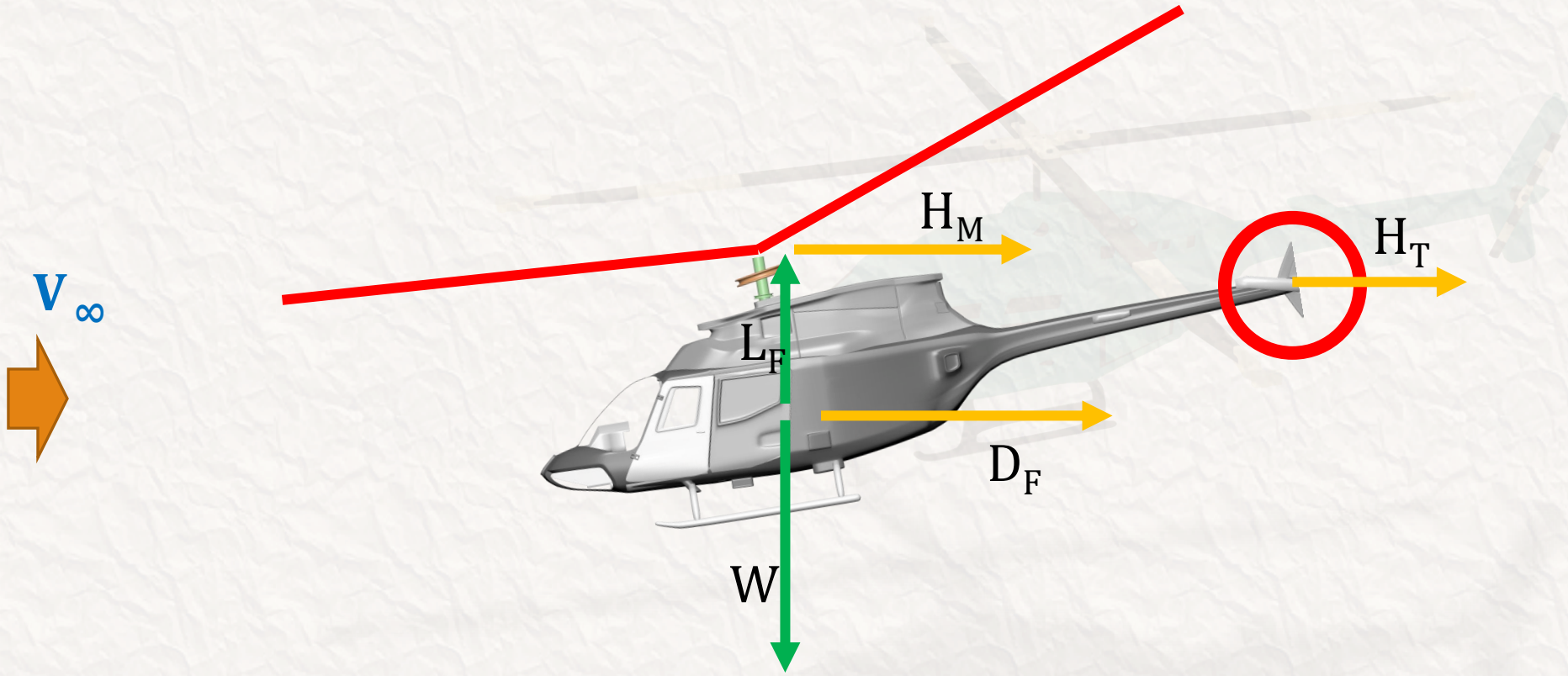
Trim Conditions for helicopter steady level forward flight

Meaning of Trim

- State of the vehicle and the control inputs necessary for a given flight condition

- Coupled and non-linear system: Iterative approach.

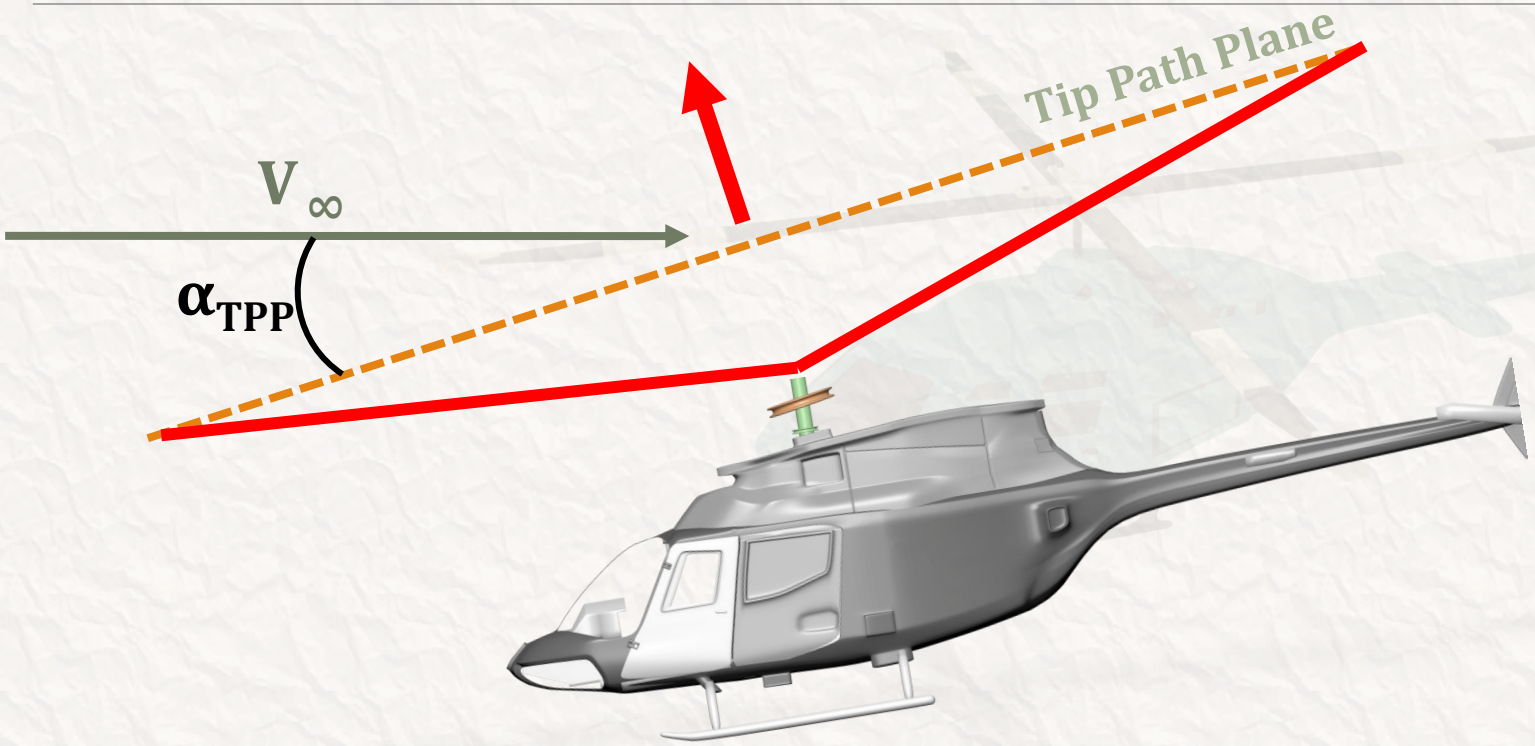
Fuselage Forces



Drag Forces = Fuselage drag (D_F) + Main rotor H-force (H_M) + Tail rotor H-force (H_T)

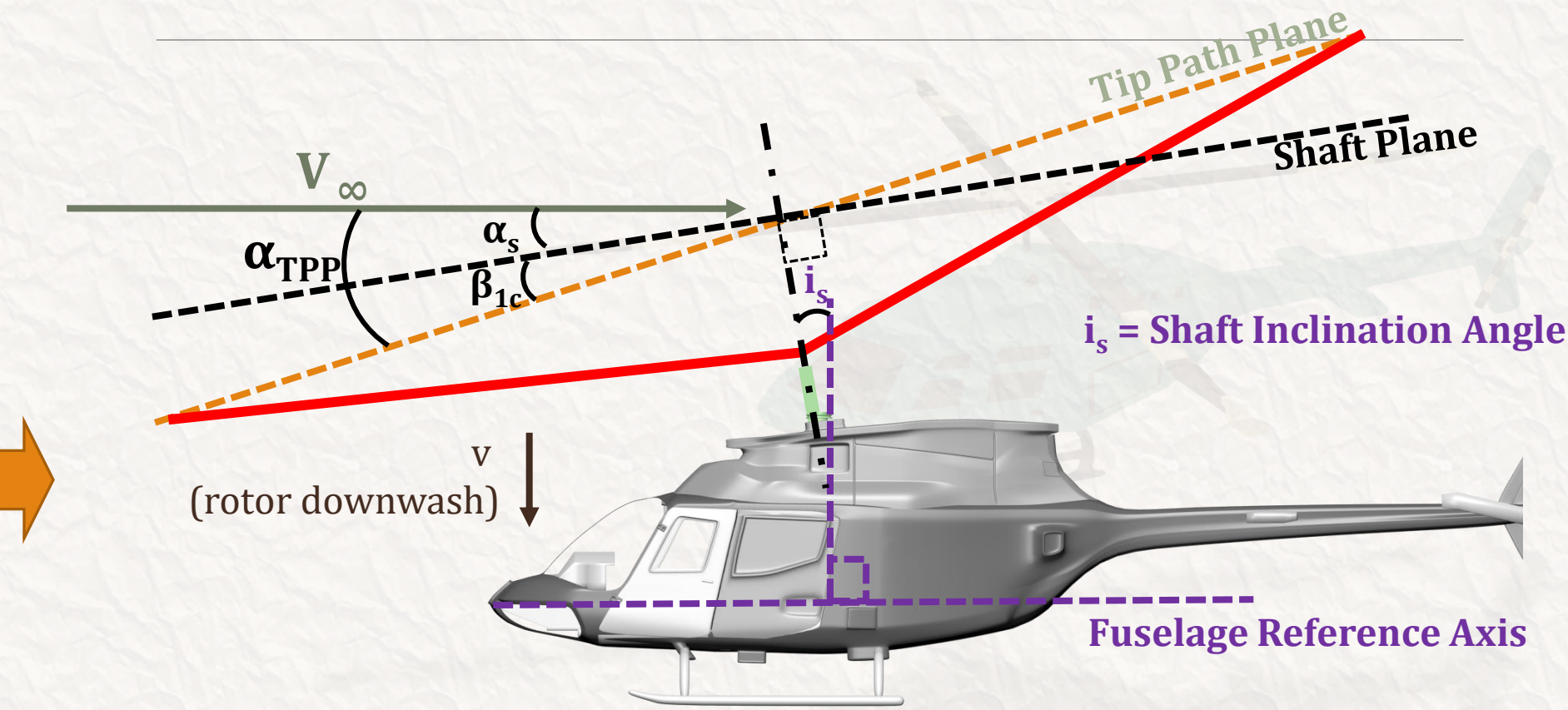
Vertical Forces = Weight (W) – Fuselage Lift (L_F)

TPP Angle



- $T\cos\alpha_{TPP} = GW - L_F$
- $T\sin\alpha_{TPP} = D_F + H_M + H_T$
- $\alpha_{TPP} = \tan^{-1}\left[\frac{D_F+H_M+H_T}{GW-L_F}\right]$

Angle of Attack of TPP, Shaft, Fuselage



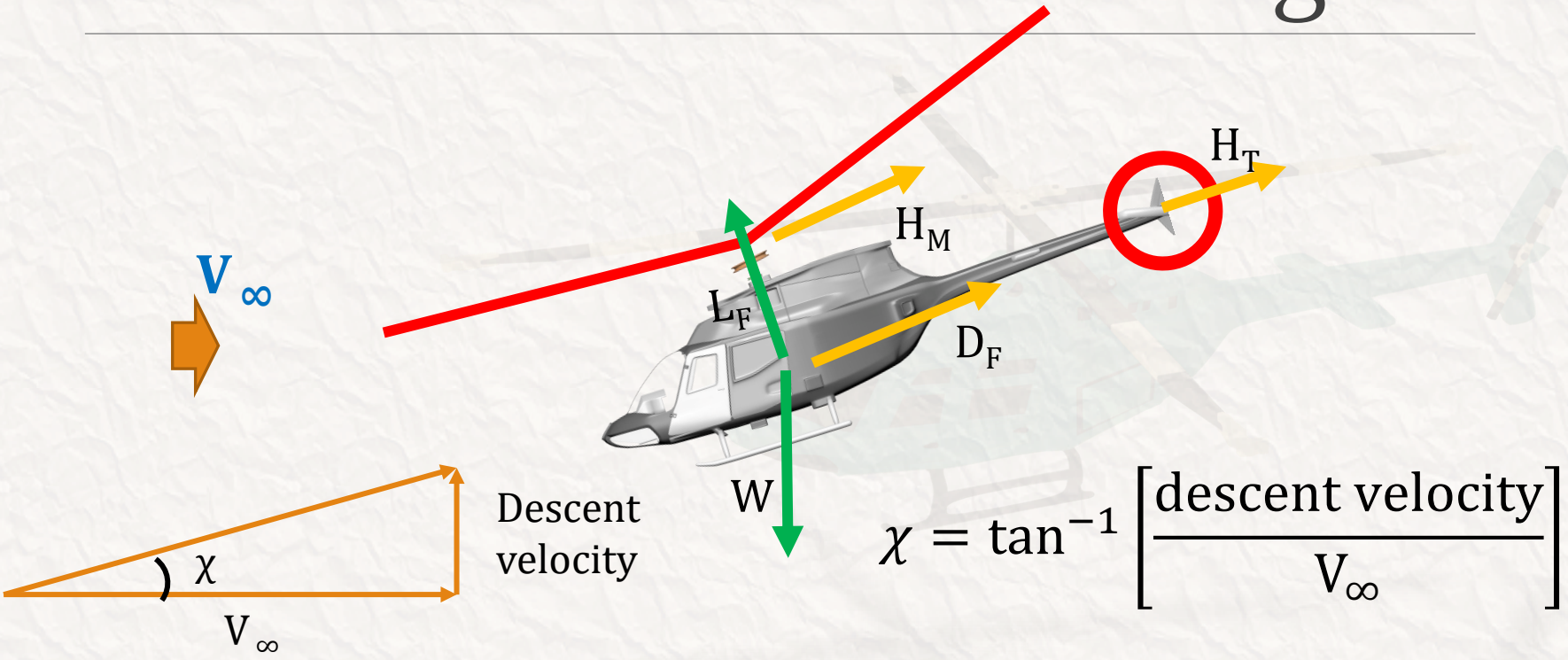
$$\alpha_{TPP} = \alpha_s + \beta_{1c}; \qquad \alpha_F = \alpha_s - i_s + v/V_\infty$$

$$\boldsymbol{\alpha_F = \alpha_{TPP} - \beta_{1c} - i_s + v/V_\infty}$$

Iteration Loop

1. Assume initial $\alpha_F = 0$
2. Find L_F and D_F from wind tunnel tables of the fuselage
3. Compute T and α_{TPP} using drag, weight, fuselage forces, H_M, and H_T. Neglect H_M and H_T in first iteration
4. Use information on c.g of fuselage to find α_F
5. Iterate step 2 to 4 until quantities converge.

Autorotation in forward flight



- $T\cos\alpha_{TPP} = W\cos\chi - L_F$
- $T\sin\alpha_{TPP} = D_F + H_M + H_T - W\sin\chi$
- $\alpha_{TPP} = \tan^{-1}\left[\frac{D_F + H_M + H_T - W\sin\chi}{W\cos\chi - L_F}\right]$

Iteration Loop

1. Assume an initial descent velocity
2. Use the iterative procedure to get α_F, forces, etc. as described earlier
3. Compute required power: Main rotor power + Tail rotor power + Transmission losses
4. Compute available power = W * Descent velocity
5. Equate available power and required power by adjusting the descent velocity
6. Iterate step 2 to 5 until power needed = power available.